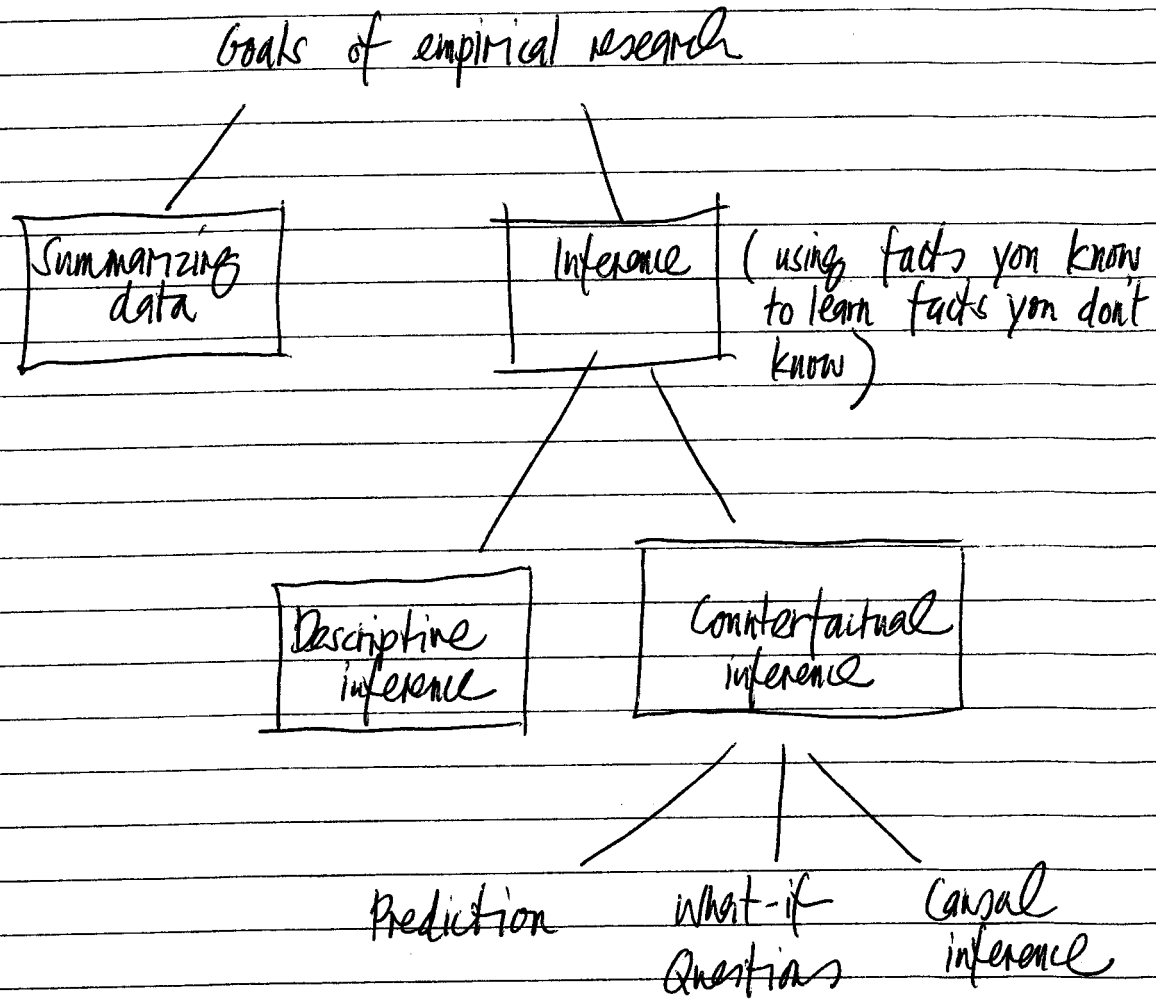
UP 504 **Causal Inference** (Intro)

Target quantities of interest: Inference vs. summarisation



Equivalent linear regression notation

* Standard version:

systematic stochastic

$$Y_i = m(\underline{x}) + e = x_i\beta + e_i$$

$$e_i \sim f_N(e_i \mid 0, \sigma^2)$$

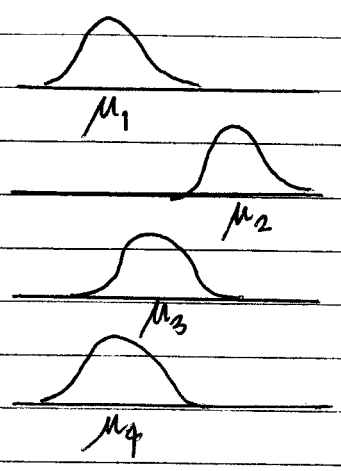
Stochastic (from στόχος ≡ aim, guess); process that is fundamentally non-deterministic, ie contains a random component

* Alternative version:

$$Y_i \sim f_N(y_i \mid \mu_i, \sigma^2) \quad \text{stochastic}$$

$$\mu_i = x_i\beta \quad \text{systematic}$$

Understanding the alternative regression notation:



Is a histogram of Y a test for normality?

Generalised alternative notation for most statistical models

$$Y_i \sim f(y_i \mid \theta_i, \alpha) \qquad \text{stochastic}$$

$$\theta_i = g(X_i, \beta) \qquad \text{systematic}$$

where Y_i random outcome variable

y_i realization of Y_i

$f(\cdot)$ probability density function

θ_i a systematic feature of the density that varies over observations i

$g(\cdot)$ functional form (linear, exponential ...)

X_i explanatory variables

β effect parameters

α ancillary parameter (feature of density that remains constant over i)

Forms of uncertainty

$$Y_i \sim f(y_i \mid \theta_i, \alpha) \qquad \text{stochastic}$$
$$\theta_i = g(X_i, \beta) \qquad \text{systematic}$$

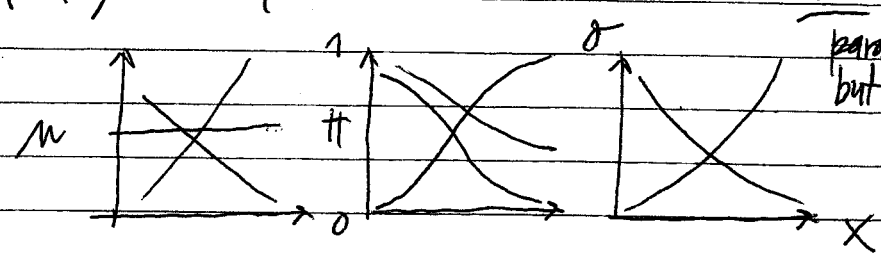
* **Estimation uncertainty**: Lack of knowledge of β and α. Vanishes as n gets larger

* **Fundamental uncertainty**: Represented by the stochastic component. Exists no matter what the researcher does; no matter how large n is.

↳ Systematic component examples:

$$E(Y_i) \equiv \mu_i = X\beta = \beta_0 + \beta_1 X_{1i} + \dots \beta_k X_{ki}$$
$$Pr(Y_i = 1) \equiv \pi_i = 1/[1 + e^{-x_i\beta}]$$
$$V(Y_i) \equiv \sigma_i^2 = e^{x_i\beta}$$

NB: "β" is an effect parameter in each, but meanings differ.



* Heckman et al (2010)
IZA working Papers

UP504 **The experimental ideal**

Perry pre-school randomized experiment (1962) — HighScope early childhood education approach

(i) 123 black pre-schoolers in Ypsilanti, MI
(ii) receive intensive intervention pre-school, home visits
(iii) Follow-up data until 1993 (from age 3 to age 40)

↳ basis for Head Start program; new evidence* that program worked for females, less-measurable effects for males

↳ ~~Evaluation → Agenda → Formulation~~ → Adoption → Intervention

*** Selection problem** (from correlation to causal inference)

From J.P Harding et. al. (JUE, 2012)

		Purchase	Sale	Hold. P.	Ann %	Std. Err.
REO	900	$136k	$246k	7.3	10.4 %	(12.8%)
Conventional	5,335	$173k	$232k	5.9	6.5 %	(9.5%)

Group	Sample	Mean Health	Std. Err
Hospital	7,774	3.21	0.014
No Hospital	90,049	3.93	0.003

use also Harding et al. (JUE, 2012) Foreclosure myth?

$$\text{t-stat} = \frac{\mu_1 - \mu_2}{\sqrt{\sigma_1^2/N_1 + \sigma_2^2/N_2}} = 58.9$$

Foreclosed vs. non-foreclosed houses

↳ Add. "Moving to Opportunity"

Specify treatment as binary variable

$D_i = \{0, 1\}$ → potential outcome, irrespective of whether treatment occurs

∴ Two problems with observational data: (i) relationship only holds with certain probability; (ii) observe actual states only

Potential outcome $\begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$

Causal effect $\equiv Y_{1i} - Y_{0i}$ (Rubin causal model; Holland (1986))

$$Y_i = \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$$

$$= Y_{0i} + \underbrace{(Y_{1i} - Y_{0i})}_{\text{CAUSAL EFFECT}} D_i$$

↳ Both potential outcomes are never observed for the same person ∴ must rely on some population distributions and compare averages:

$$\overbrace{E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]}^{\text{observed difference in average health}} = \overbrace{\{E[Y_{1i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 1]\}}^{\text{PARAMETER OF INTEREST: ~~Average treatment~~ effect on the treated}}$$

→ mean outcome of participants, had they not particip.

$$+ \underbrace{\{E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]\}}_{\text{selection bias}}$$

→ mean outcome of nonparticipants cannot be used as proxy

$$E[Y_{1i} - Y_{0i} \mid D_i = 1]$$ AVERAGE CAUSAL EFFECT OF TREATMENT

∴ selection bias is negative in this example and can completely mask treatment effect.

Selection bias arises because Y_{0i} depends on D_i (dependence of Y_{0i} and D_i)

Random assignment solves selection bias

* D_i is rendered independent of potential outcome

due to independence of Y_{0i} and D_i

RANDOMIZATION

$$E[Y_i|D_i=1] - E[Y_i|D_i=0] =$$

$$= E[Y_{1i}|D_i=1] - E[Y_{0i}|D_i=0]$$

$$= E[Y_{1i}|D_i=1] - E[Y_{0i}|D_i=1]$$

$$\therefore E[Y_{1i}|D_i=1] - E[Y_{0i}|D_i=1] =$$

$$= E[Y_{1i} - Y_{0i} | D_i=1] = E[Y_{1i} - Y_{0i}]$$

↳ Studying the impact of policy

Alternatives: (i) full general equilibrium analysis

(ii) partial equilibrium structural research

Let $D^*=1$ for persons who participate in program with random assignment ($r=1$ or $r=0$)

$D^*=0$ for everybody else

$$\therefore E[Y_{1i}|D_i=1] = E[Y_{1i}|r=1 \text{ and } D^*=1] \text{ and}$$
$$E[Y_{0i}|D_i=1] = E[Y_{1i}|r=0 \text{ and } D^*=1]$$

NB: Random assignment does not remove selection bias, but balances the bias between the participant and non-participant samples.

Assume same treatment effect for all:

$$Y_{1i} - Y_{0i} = \beta \quad \text{and recall that } Y_i = Y_{0i} + (Y_{1i} - Y_{0i}) D_i$$

$$\circledast \quad Y_i = E[Y_{0i}] + [Y_{1i} - Y_{0i}] D_i + [Y_{0i} - E[Y_{0i}]]$$

$$= \alpha + \beta D_i + \eta_i$$

$$\therefore \; E[Y_i \mid D_i = 1] = \alpha + \beta + E[\eta_i \mid D_i = 1]$$

$$E[Y_i \mid D_i = 0] = \alpha + E[\eta_i \mid D_i = 0]$$

↳ such that $E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] =$

$$\underbrace{\beta}_{\text{Treatment effect}} + \underbrace{E[\eta_i \mid D_i = 1] - E[\eta_i \mid D_i = 0]}_{\text{selection bias}}$$

↳ The selection bias amounts to correlation between the regression error η_i and the regressor D_i

Since,

$$E[\eta_i \mid D_i = 1] - E[\eta_i \mid D_i = 0] =$$

$$E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$$

$\therefore$ this correlation between η_i and D_i reflects the <u>difference in the no-treatment potential outcomes</u> between those who get treated and those who don't.

Notice that under randomization:

$$E[Y_i \mid r = 1 \text{ and } D_i^* = 1] = \alpha + \beta + E[\eta_i \mid D_i^* = 1]$$

$$E[Y_i \mid r = 0 \text{ and } D_i^* = 1] = \alpha + E[\eta_i \mid D_i^* = 1]$$

$$\therefore E[Y_i \mid r = 1 \text{ and } D_i^* = 1] - E[Y_i \mid r = 0 \text{ and } D_i^* = 1] = \beta$$

* ↳ but randomization does not guarantee that the selection problem disappears (ie $E[\eta_i \mid D_i^* = 1] \neq 0$); but with randomized data selection bias is spread across samples such that it cancels out when calculating the treatment effect.

Must be as uncorrelated as possible

→ Use <u>control variables</u> to get better "fix" on β: $Y_i = \alpha + \beta D_i + X_i' \gamma + \eta_i$

Conditional independence assumption (CIA)

(i) Experiments ensure that causal variable of interest is independent of potential outcomes

(ii) CIA provides implicit justification for causal interpretation of regression estimates

↳ since variety of control variables (all of which are assumed to be observable) are held constant

∴ if we can $\boxed{\text{"select on observables"}}$

eg. imagine X_i as vector that explains neighbourhood characteristics, ability, family background etc.

Without CIA, $E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$ will contain a selection bias. CIA means that conditional on X_i, bias disappears:

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i | X_i$$

independence, given $D_i | X_i$

$$\therefore E[Y_i | X_i, D_i = 1] - E[Y_i | X_i, D_i = 0]$$

$$= E[Y_{1i} - Y_{0i} | X_i]$$

CAUSAL INTERPRETATION OF REGRESSION ANALYSIS

Regression Analysis in R

$$TSS = ESS + RSS$$

PROOF $\sum_1^n (y_i - \bar{y})^2 = \sum^n (\hat{y}_i - \bar{y}_i)^2 + \sum^n (y_i - \hat{y}_i)^2$

↳ expand $\sum_1^n (y_i - \bar{y} + \hat{y}_i - \hat{y}_i)^2 = \sum^n (\hat{y}_i - \bar{y} + \hat{e}_i)^2$

$$= \sum \left((\hat{y}_i - \bar{y})^2 + 2\hat{e}_i(\hat{y}_i - \bar{y}) + \hat{e}_i^2 \right)$$

$$= \sum (\hat{y}_i - \bar{y})^2 + \sum e_i^2 + 2\sum \underbrace{\hat{e}_i(x'\hat{\beta} - \bar{y})}_{\Rightarrow 0 \text{ due to } x'e}$$

$$= ESS + RSS$$

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \quad \text{because} \quad \frac{ESS}{TSS} = \frac{TSS - RSS}{TSS}$$